Vandit Parekh

Mobile: +91-9157585969

Email: parekhvandit@gmail.com

Education

- **Vellore Institute of Technology, Bhopal** — Bhopal, India
  Bachelor of Technology - Computer Science — *September 2023 - May 2027*

  Courses: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Data Mining & Data Warehouse Management, AWS, Computer Networks, Operating Systems, Data Structures & Algorithms, Object-Oriented Programming.

Skills Summary

Languages: C++, Python, Java, SQL, HTML, Unix/Linux, Assembly
Platforms: Linux, Windows, AWS
Co-Curricular: Public Speaking, Technical Writing, Team Collaboration, Debate

Projects

- **Leptospirosis Risk Predictor**:
  - Engineered a full-stack web application using React, Node.js, and PostgreSQL to predict and analyze Leptospirosis risk across European nations based on climate factors like temperature and humidity.
  - Integrated health records from the ECDC with climate reanalysis data (ERA5) from ECMWF, utilizing Pandas and NumPy to preprocess and merge complex datasets for model training.
  - Trained and optimized a Random Forest machine learning model using Scikit-Learn and Joblib to generate accurate epidemiological risk predictions based on the unified dataset.
  - Designed a robust data pipeline that connects the Python-based predictive backend with the frontend interface, dynamically loading and displaying risk assessments for users.
- **Lightweight Pneumonia Detection System**:
  - Developed a computationally efficient computer vision pipeline to classify chest X-ray images into Normal and two distinct Pneumonia types, circumventing the need for resource-heavy deep learning architectures.
  - Extracted handcrafted features using Canny edge detection and Histogram of Oriented Gradients (HOG) to accurately capture critical structural and textural information from medical scans.
  - Implemented and trained a Support Vector Machine (SVM) classifier using Scikit-Learn to process the extracted features, delivering a lightweight and highly responsive diagnostic tool.
  - Architected a modular Python codebase with distinct training and inference scripts, streamlining model generation and ensuring rapid, scalable testing of new X-ray images.
- **EPICS Telehealth Platform**:
  - Engineered a comprehensive telemedicine platform utilizing Flask and MySQL to connect remote patients with global specialists, facilitating secure, longitudinal medical history tracking and collaborative case management.
  - Integrated WebRTC and Flask-SocketIO to enable robust, real-time video consultations, optimizing peer-to-peer communication for reliable healthcare delivery in low-bandwidth environments.
  - Developed an AI-driven specialist recommendation engine powered by the Gemini API, designed to dynamically match patients with doctors based on clinical symptoms, vital trends, geographical proximity, and budget.
  - Implemented proactive health analytics and safety modules that track long-term vital trends for predictive care and automatically cross-reference historical allergy data to prevent adverse prescription reactions.

Achievements

- **National Top 10 Finalist (NISM Ideathon):** Architected a conceptual proposal (Project AEGIS) for a proactive, technology-driven investor protection system, projecting a 60-75% reduction in financial scams upon full regulatory enforcement.
- **HealthHack Hackathon (Top 50):** Secured a top 50 rank among 850+ teams, showcasing strong problem-solving, algorithmic thinking, and efficient full-stack coding under pressure.
- **NPTEL Top 5% Performer (IIT Kharagpur):** Recognized for exceptional academic performance and technical understanding in an advanced Internet of Things (IoT) certification course.
- **Chief of Operations (VITBMUN):** Oversaw organizational logistics, managed finances, facilitated high-level discussions, and led the core organizing team for a large-scale collegiate event.
- **Silver Medalist (Inter-College Football):** Demonstrated strong team collaboration, strategic discipline, and physical endurance in a highly competitive tournament.